

CONCEPT QUESTIONS – CH : 5

Why Numerical Methods?

5-1C

With powerful computers and software packages readily available, do you think obtaining analytical solutions to engineering problems will eventually disappear from engineering curricula?

No, analytical solutions will not disappear since analytical solutions are used engineers understand principles and provide exact results under simplified assumptions. Analytical solutions are also used as a benchmark for checking numerical results while numerical methods can handle complex and realistic applications.

5-2C

What are the limitations of the analytical solution methods?

Analytical solution methods are limited because they typically require simple geometries, idealized boundary conditions, and simplifying assumptions which is not accurate in real heat transfer problems.

5-3C

Consider a heat conduction problem that can be solved both analytically, by solving the governing differential equation and applying the boundary conditions, and numerically, by a software package available on your computer. Which approach would you use to solve this problem? Explain your reasoning.

If the problem can be solved analytically it should be used as it will provide an exact solution under the given assumption. However, I would also use numerical solutions to verify the results or where the assumptions are not constant.

5-4C

What is the basis of the energy balance method? How does it differ from the formal finite difference method? For a specified nodal network, will these two methods result in the same or a different set of equations?

The energy balance method is based on applying conservation of energy to a small control volume around each node. The heat entering, leaving, and being generated within the element are balanced to form an equation. The formal finite difference method starts with a differential equation and replaces derivatives with finite difference approximations. For a nodal network both methods result in the same set of algebraic equations, even though they are derived differently.

5-5C

How do numerical solution methods differ from analytical ones? What are the advantages and disadvantages of numerical and analytical methods?

Numerical methods differ from analytical methods as they provide approximate solutions by discretizing the domain, while analytical methods provide exact solutions under simplified assumptions. Analytical methods offer clear insight and exact results but are limited by the geometry and boundary conditions. Numerical methods can handle complex and realistic problems, but they introduce round-off errors due to discretization.

5-6C

Two engineers are to solve an actual heat transfer problem in a manufacturing facility. Engineer A makes the necessary simplifying assumptions and solves the problem analytically, while engineer B solves it numerically using a powerful software package. Engineer A claims he solved the problem exactly and thus his results are better, while engineer B claims that he used a more realistic model and thus his results are better. To resolve the dispute, you are asked to solve the problem experimentally in a lab. Which engineer do you think the experiments will prove right? Explain.

The experiments will likely support Engineer B results since numerical solutions provide a more realistic model with fewer assumptions. Engineer A may be exact, but it used a simplified model. Real systems require more accuracy and therefore the model would represent reality more.

Finite Difference Formulation of Differential Equations

5-7C

Define these terms used in the finite difference formulation: node, nodal network, volume element, nodal spacing, and difference equation.

Node: A specific point in the domain where the temperature is calculated.

Nodal Network: The entire grid/mesh of nodes that represent the physical region being analyzed.

Volume Element: A small control volume surrounding a node where energy balance is applied.

Nodal Spacing: The distance between adjacent nodes in the grid/mesh.

Difference Equations: The algebraic equation obtained by replacing derivatives with finite differences, relating the temperature at a node to neighboring nodes.

One-Dimensional Steady Heat Conduction

5-13C

Explain how the finite difference form of a heat conduction problem is obtained by the energy balance method.

The finite difference form is obtained by applying an energy balance to a small control volume around each node. The rates of heat conduction into and out of the volume, along with any internal heat generation, are accounted for using conservation of energy.

The heat transfer rates are then expressed in terms of temperature differences between neighboring nodes, resulting in an algebraic equation that relates the temperature at a node to the surrounding.

5-14C

What are the basic steps involved in solving a system of equations with Gauss-Seidel method?

1. Write the finite difference equation explicitly for each node.
2. Make a reasonable initial guess for each unknown nodal temperature
3. Use explicit equations to calculate new values for each nodal temperature
4. Repeat the process until convergence within some tolerable error is achieved

5-15C

Consider a medium in which the finite difference formulation of a general interior node is given in its simplest form as

$$\frac{T_{m-1} - T_m + T_{m+1}}{\Delta x^2} + \frac{\dot{q}}{k} = 0$$

1. Is heat transfer in this medium steady or transient?
2. Is heat transfer one-, two-, or three-dimensional?
3. Is there heat generation in the medium?
4. Is the nodal spacing constant or variable?
5. Is the thermal conductivity of the medium constant or variable?

1. Steady, since there is no time dependent term present.
2. One-dimensional, since the temperature are related only in one spatial direction.
3. Yes, since a heat generation term appears in the equation.
4. Constant, since a single uniform spacing term is used.
5. Constant, since it is not expressed as varying with position.

5-16C

How is an insulated boundary handled in finite difference formulation of a problem? How does a symmetry line differ from an insulated boundary in the finite difference formulation?

An insulated boundary is handled as a zero-heat flux boundary. In finite differences treat the boundary as a “mirror” so the boundary node can be handled like an interior node using the mirror image concept.

A symmetry line also implies zero heat flux across the line. The finite difference for a symmetry line is the same as an insulated boundary. Insulation is a real boundary, while symmetry is an internal boundary.

5-17C

How can a node on an insulated boundary be treated as an interior node in the finite difference formulation of a plane wall? Explain.

A node on an insulated boundary can be treated as a mirror-image concept. Since an insulated boundary implies zero heat flux, the temperature gradient at the boundary is zero. Therefore, the boundary node is the same as an interior node.

5-18C

In the energy balance formulation of the finite difference method, it is recommended that all heat transfer at the boundaries of the volume element be assumed to be into the volume element even for steady heat conduction. Is this a valid recommendation even though it seems to violate the conservation of energy principle?

Yes, this is valid since the energy balance method assumes all heat transfer is into the control volume. The direction of heat flow is accounted for automatically due to the temperature difference. Even for steady conduction, conservation of energy is not violated since the algebraic sum of all heat transfer will equal zero at steady state.

Two-Dimensional Steady Heat Conduction

5-54C

What is an irregular boundary? What is a practical way of handling irregular boundary surfaces with the finite difference method?

An irregular boundary is a boundary that does not align neatly with a uniform nodal grid/mesh pattern used in finite difference methods. For example, a curved or angled surface. A practical way to handle irregular boundaries is to approximate the boundary using a grid that closely follows the shape. In practice the irregular shape is often approximated as a stepped or piecewise representation that fits the grid.

5-55C

Consider a medium in which the finite difference formulation of a general interior node is given in its simplest form as the following:

$$T_{node} = \frac{T_{left} + T_{top} + T_{right} + T_{bottom}}{4}$$

1. Is heat transfer in this medium steady or transient?
2. Is heat transfer one-, two-, or three-dimensional?
3. Is there heat generation in the medium?
4. Is the nodal spacing constant or variable?
5. Is the thermal conductivity of the medium constant or variable?

1. Steady, since there is no time dependent term.
2. Two-dimensional, since temperature depends on left, right, top, and bottom neighbors.
3. No, since there is no heat generation term
4. Constant, since the node temperature is the average of four surrounding nodes.
5. Constant, since no variation in conductivity appears in the formula.

5-56C

Consider a medium in which the finite difference formulation of a general interior node is given in its simplest form as the following:

$$T_{left} - T_{top} + T_{right} + T_{bottom} - 4T_{node} + \frac{\dot{q}_{node}l^2}{k} = 0$$

1. Is heat transfer in this medium steady or transient?
2. Is heat transfer one-, two-, or three-dimensional?
3. Is there heat generation in the medium?
4. Is the nodal spacing constant or variable?
5. Is the thermal conductivity of the medium constant or variable?

1. Steady, since there is no time dependent term.
2. Two-dimensional, since temperature depends on left, right, top, and bottom neighbors.
3. Yes, since a heat generation term appears in the equation.
4. Constant, since a single-spacing term is used
5. Constant, since it appears as a single value and is not expressed as varying with position.

Transient Heat Conduction

5-85C

How does the finite difference formulation of a transient heat conduction problem differ from that of a steady heat conduction problem? What does the term $pA\Delta x C_p \frac{T_m^{i+1} - T_m^i}{\Delta t}$ represent in the transient finite difference formulation?

In steady formulation, the finite difference equation is just a balance of heat transfer terms at a node. In transient formulation, the equation includes an additional time dependent storage term, since the node temperature changes with time. The term $pA\Delta x C_p \frac{T_m^{i+1} - T_m^i}{\Delta t}$ represents the rate of change of thermal energy stored in the control volume around the node.

5-86C

**What are the two basic methods of solution of transient problems based on finite differencing?
How do heat transfer terms in the energy balance formulation differ in the two methods?**

The two basic methods are explicit and implicit methods. The explicit method evaluates heat transfer terms using temperatures at the current time, and future temperatures are computed directly. The implicit method writes the heat transfer terms of temperature at the future time level, which results in a system of simultaneous equations that must be solve at each time step.

5-87C

The explicit finite difference formulation of a general interior node for transient heat conduction in a plane wall is given by the following:

$$T_{m-1}^i - 2T_m^i + T_{m+1}^i + \frac{\dot{q}_m^i \Delta x^2}{k} = \frac{T_m^{i+1} - T_m^i}{\tau}$$

Obtain the finite difference formulation for the steady case by simplifying the relation above.

For the steady case, the temperature does not change with time. Therefore, the time dependent term becomes zero. When the time terms is removed the equation reduces to the steady-state finite difference form:

$$T_{m-1}^i - 2T_m^i + T_{m+1}^i + \frac{\dot{q}_m^i \Delta x^2}{k} = 0$$

5-88C

Consider transient one-dimensional heat conduction in a plane wall that is to be solved by the explicit method. If both sides of the wall are at specified temperatures, express the stability criterion for this problem in its simplest form.

For a plane wall divided into nodes with a spatial step Δx and a time step Δt , the governing equation for an interior node m at the new time step $i+1$ is:

$$T_m^{i+1} = Fo(T_{m-1}^i + T_{m+1}^i) + (1 - 2Fo)T_m^i$$

For the explicit method in 1-D transient conduction, stability requires that the Fourier number satisfies:

$$Fo = \frac{\alpha \Delta t}{(\Delta x^2)} \leq \frac{1}{2}$$

The expression ensures that the numerical solution remains stable and does not diverge.

5-89C

Consider transient one-dimensional heat conduction in a plane wall that is to be solved by the explicit method. If both sides of the wall are subjected to specified heat flux, express the stability criterion for this problem in its simplest form.

For a plane wall with heat flux, we focus on the energy balance at the surface node. Using the explicit method, the temperature at the boundary node T_0 at the next time step $i+1$ is determined by:

$$T_0^{i+1} = 2Fo \left(T_1^i + \frac{\dot{q}_s \Delta x}{k} \right) + (1 - 2Fo)T_0^i$$

For the explicit method, the stability requirement is still based on keeping the primary coefficient nonnegative. For 1-D transient conduction, the simplest stability criterion is:

$$Fo = \frac{\alpha \Delta t}{(\Delta x^2)} \leq \frac{1}{2}$$

The expression ensures that the numerical solution remains stable and does not diverge.

5-90C

The explicit finite difference formulation of a general interior node for transient two-dimensional heat conduction is given by the following:

$$T_{node}^{i+1} = \tau(T_{left}^i + T_{top}^i + T_{right}^i + T_{bottom}^i) + (1 - 4\tau)T_{node}^i + \tau \frac{\dot{q}_{node}^i l^2}{k}$$

Obtain the finite difference formulation for the steady case by simplifying the relation above

For steady case, temperatures do not change with time so:

$$T_{node}^{i+1} = T_{node}^i = T_{node}$$

Substituting this into the transient two-dimensional heat conduction:

$$T_{left} - T_{top} + T_{right} + T_{bottom} - 4T_{node} + \frac{\dot{q}_{node} l^2}{k} = 0$$

5-91C

Is there any limitation on the size of the time step Δt in the solution of transient heat conduction problems using each of the following methods:

1. the explicit method.
2. the implicit method

1. Yes, since the time step must be small to satisfy the stability criterion.
2. No! since there is no stability limitation, so Δt can be used from a stability standpoint.

5-92C

Express the general stability criterion for the explicit method of solution of transient heat conduction problems.

For the explicit method, stability requires that the coefficient in the finite difference equation remain greater than or equal to zero. The general form is the sum of the Fourier number in all spatial direction must be less than or equal to 1/2.

$$T_m^{i+1} = A \cdot T_{m-1}^i + B \cdot T_m^i + C \cdot T_{m+1}^i + \dots$$

$$\sum Fo = \sum \frac{\alpha \Delta t}{(\Delta x^2)} \leq \frac{1}{2}$$

This ensures that the numerical solution remains stable and does not diverge.

5-93C

Consider transient two-dimensional heat conduction in a rectangular region that is to be solved by the explicit method. If all boundaries of the region are either insulated or at specified temperatures, express the stability criterion for this problem in its simplest form.

For two-dimensional transient conduction using the explicit method with uniform grid spacing, stability requires:

$$T_{m,n}^{i+1} = Fo(T_{m+1,n}^i + T_{m-1,n}^i + T_{m,n+1}^i + T_{m,n-1}^i) + (1 - 4Fo)T_{m,n}^i$$

Where a uniform rectangular grid with equal spacing in both directions ($\Delta x = \Delta y$). If the spacing is equal in both directions, then:

$$2Fo = 2 \frac{\alpha \Delta t}{(\Delta x^2)} \leq \frac{1}{2}$$

Which simplifies to:

$$Fo = \frac{\alpha \Delta t}{(\Delta x^2)} \leq \frac{1}{4}$$

This is the simplest stability criterion for 2-D transient explicit conduction with insulated or specified-temperature boundaries.

5-94C

The implicit method is unconditionally stable and thus any value of time step Δt can be used in the solution of transient heat conduction problems. To minimize the computation time, someone suggests using a very large value of Δt since there is no danger of instability. Do you agree with this suggestion? Explain.

No, since a large time step will produce poor accuracy even though implicit method is unconditionally stable. When a large time step is used important transient behavior can be missed. Stability just means that it won't cause issues numerically, it does not mean the solution is accurate. The time step should be chosen based on the time scale of the physics that need to be captured.

Special Topic: Controlling the Numerical Error

5-121C

Why do the results obtained using a numerical method differ from the exact results obtained analytically? What are the causes of this difference?

Numerical methods differ from exact results since they produce approximation. The main cause for the difference is discretization error which is replacing derivatives/continuous domain with finite differences on a grid. The second cause is round off error which introduces small numerical errors due to finite precision.

5-122C

What is the cause of the discretization error? How does the global discretization error differ from the local discretization error?

Discretization errors are caused by the approximations used in the formulation of the numerical method. Furthermore, Discretization errors are caused by replacing the derivative with finite difference approximations and representing a continuous domain with a finite grid/mesh which introduces truncation error. The local discretization error is the error made in a single step or at a single node due to the approximation. The global discretization error is the accumulated effect of the local errors over the entire domain.

5-123C

Can the global (accumulated) discretization error be less than the local error during a step? Explain.

Yes. Although the global error results from the accumulation of local errors, the local errors can cancel each other due to opposite signs. Due to the cancellation, the global discretization error can sometimes be smaller than the local error at a given step.

5-124C

How is the finite difference formulation for the first derivative related to the Taylor series expansion of the solution function?

The finite difference formulation for the first derivative is obtained by expanding the temperature function in a Taylor series about a point and then neglecting the higher order terms. Rearranging the Taylor series expansion a finite difference approximation for the derivative is derived. The higher order terms that are neglected is what causes discretization error.

5-125C

Explain why the local discretization error of the finite difference method is proportional to the square of the step size. Also explain why the global discretization error is proportional to the step size itself.

The local discretization error is proportional to the square of the step size because of the finite difference approximation that is derived from a Taylor series expansion in which the first neglected term is of order (Δx^2).

The global discretization error is proportional to the step size itself because the local errors accumulate over many steps or nodes. As the second-order local errors build up across the domain, the overall error becomes first order proportional to the step size.

5-126C

What causes the round-off error? What kind of calculations are most susceptible to round-off error?

Round-off errors are caused by the computer's use of a limited number of significant digits and continuously rounding (or chopping) off the digits it cannot retain. Furthermore, Round-off error is caused by the finite precision with which numbers are stored and manipulated in a computer. Since a computer cannot represent most real numbers exactly, small rounding errors occur during calculations.

Calculations that involve subtracting nearly equal numbers or performing many repetitive computations are most susceptible to round-off errors because small errors can accumulate or become magnified.

5-127C

What happens to the discretization and the round-off errors as the step size is decreased?

As the step size decreases, the discretization error decreases because the finite difference approximation more closely represents the continuous system. However, the round-off error increases, since reducing the step size increases the number of calculations and iteration, allowing round-off errors to accumulate.

5-128C

Suggest some practical ways of reducing the round-off error.

Some practical ways to reduce round-off error is by high precision arithmetic, avoid subtraction of equal numbers, minimize loss of significant digits, and reduce the total number of unnecessary calculations and iterations.

5-129C

What is a practical way of checking if the round-off error has been significant in calculations

A practical way of checking if the round-off error is significant in calculations is to repeat calculations using higher precision and compare the results. If the results change noticeably, round-off errors are significant. If they are close to the same, then the round-off error is negligible.

5-130C

What is a practical way of checking if the discretization error has been significant in calculations?

A practical way of checking if the discretization error has been significant in calculations is to refine the grid and reduce the step size and repeat the calculation. If the solution changes noticeably as the grid is refined, discretization error was significant. If the solution is close to the same, then the discretization error is small and therefore the solution is grid independent.